# Milestone 1

*Is Lausanne's housing stock evolving to match its changing social fabric?*

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## 1. Dataset

Our project draws on six official datasets published by the City of Lausanne's statistical office (Statistique Lausanne). All datasets are structured as annual tables broken down by city district (quartier), making them well-suited for both temporal and spatial analysis. The data is clean, consistently formatted, and requires minimal preprocessing — primarily reshaping from wide to long format and harmonising column names across tables.

The six datasets are:

- Population by sex and origin (Swiss / foreign, male / female): 1979–2025
- Population by age class (5 categories): 1979–2025
- Private households by size (1 to 6+ persons): 2012–2025
- Active employment by sex (establishments and full-time equivalents): 2011–2023
- Dwellings by floor surface area (7 categories, from $< 40\text{m}^2$ to $160\text{m}^2+$): 2011–2025
- Dwellings by number of rooms (1 to 6+ rooms): 2011–2025

Data quality is high: no significant missing values have been identified, units and categories are consistent across years, and the official source eliminates reliability concerns. The main preprocessing steps will be pivoting wide-format annual tables into tidy long-format structures, joining tables on year and district, and computing derived ratios (e.g. average household size, share of small apartments). We estimate 1–2 days of cleaning work in total.

## 2. Problematic

Lausanne is a city in transformation. Over the past four decades it has grown significantly, absorbed large waves of international residents — today nearly half the population is foreign-born — and seen its household composition shift markedly toward smaller, single-person living arrangements. At the same time, the city faces intense housing pressure, with vacancy rates consistently below 1%.

Our central question is: Is Lausanne's housing stock evolving to match its changing social fabric?

The visualization explores this through three interlocking lenses. First, who lives in Lausanne: how has the balance between Swiss and foreign residents shifted, and how is the age structure evolving? Second, how people live: are households getting smaller, and is rising female employment linked to greater household independence? Third, what the housing supply offers: are smaller apartments multiplying fast enough to serve a population of singles and couples, or does the stock remain oriented toward larger family units?

The target audience is twofold: the general public — residents curious about their city's trajectory — and urban planners and policymakers who need to understand whether housing production is aligned with demographic demand. The visualization aims to be accessible to both, using progressive disclosure to let casual readers grasp the headline trends while offering deeper detail for expert users.

## 3. Exploratory Data Analysis

Initial inspection of the datasets reveals several compelling patterns. On population: Lausanne's total population grew from approximately 128,000 in 1979 to over 151,000 by 2025, with foreign residents rising from roughly 23% to around 45% of the total. The age structure shows a relative increase in the [25,44] bracket, consistent with the city's role as a university and innovation hub, while the share of over-65s has grown modestly.

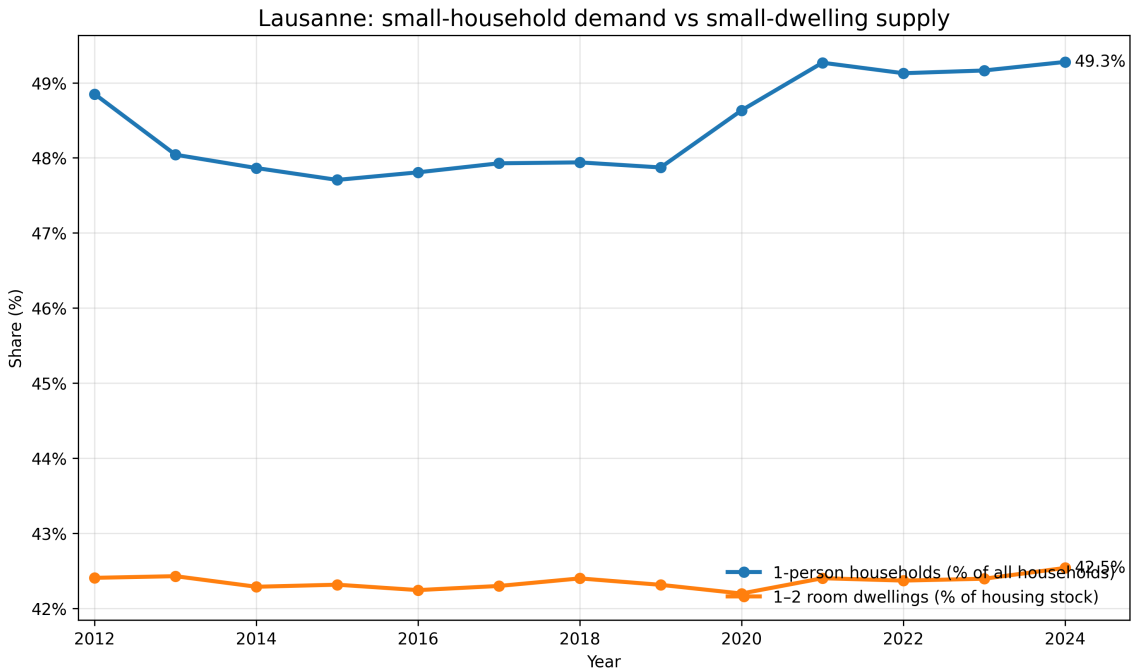


Figure 1: Comparison between the share of one-person households and the share of 1–2 room dwellings in Lausanne from 2012 to 2024. The figure highlights whether the housing stock is adapting to the rise of smaller households.

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The share of one-person households remains above the share of 1–2 room dwellings, suggesting that the housing stock has adapted more slowly than household structure.

On households: the number of private households increased faster than population, driven primarily by a surge in single-person households. By 2025, households of one person represent approximately 49% of all private households (37,286 out of 75,537), up from lower shares in 2012. Mean household size stands at 1.94 persons, confirming that Lausanne is a city of predominantly small households.

On housing stock: the dwelling distribution by number of rooms (2024) shows that 3-room apartments are the most common category (26,416 units), followed by 2-room (22,376) and

4-room (13,489). Apartments of 1–2 rooms total 35,018 units — barely enough to house the 37,286 single-person households already recorded. By surface, 14.5% of dwellings are under 40m$^2$, while 32.8% exceed 80m$^2$ (27,053 units), pointing to a structural mismatch between a predominantly small-household population and a stock still weighted toward larger apartments. Between 2010 and 2024 the total stock grew from 73,209 to 82,321 dwellings (+12.4%), but the distribution across room categories shifted only modestly.

On employment: total jobs in Lausanne reached approximately 132,000 in 2023, with women representing 52% of headcount employment but only 48% of full-time equivalents, pointing to a significant part-time differential that may influence household formation patterns.

Preprocessing needs are modest: wide-to-long pivots, year alignment across datasets (the population series begins in 1979 while housing and household data start in 2011–2012), and normalisation of district codes for any spatial layer we add.

## 4. Related Work

The Ville de Lausanne's own statistical publications (Lausanne en chiffres, annual) present these datasets as static tables and simple bar charts. The official portal offers no interactive exploration, no cross-dataset comparison, and no visualisation of the relationship between demographic and housing trends. Our work builds directly on this raw material while going far beyond its presentation.

More broadly, Swiss federal statistics (OFS/BFS) publish national-level housing and demographic trends, and platforms like the Swiss Atlas of Municipalities offer some cartographic views. However, none of these zoom into the intra-city dynamics of a single urban centre over 40+ years.

Our approach is original in three ways. First, we combine demographic and housing supply data to surface a structural mismatch that neither dataset reveals alone. Second, we bring a 46-year longitudinal view to what is usually presented as a snapshot. Third, we connect household composition trends to employment data to introduce a gender angle that is absent from standard housing analyses.

For visual inspiration we draw on the work of the Neue Zürcher Zeitung's data team, which regularly produces clean, narrative-driven Swiss demographic charts; the Washington Post's population scrollytelling pieces; and the 'Apartment for one' series published by Bloomberg CityLab on the rise of solo living in European cities. None of these use Lausanne data or the supply-demand mismatch framing we propose.

All datasets: Statistique Lausanne — lausanne.ch/officiel/statistique/quartiers/tableaux-